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History of the concept of Complexity

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Problem: The notion of Complexity was invented three times simultaneously and independently. How?

AIT's fundamental idea is that the informational content of an object can be measured by its **shortest** binary representation.

This is a profound idea, that did not occur to thinkers until the mid-twentieth century.

And then, quite remarkably, three mathematicians got the idea at the same time independently!

The discovery of Kolmogorov complexity



perceived as a real science, grounded on fundamental mathematical principles.

Conclusion:

A purely algorithmic definition of information was discovered in the 1960; known as Kolmogorov complexity, it measures information by the size of the shortest binary description.

Suggested reading:

[A beautiful page about Kolmogorov's life \(in French\).](#)

Andrej Kolmogorov's life on Scholarpedia

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